Summary of agent reports on *Matrix theory reloaded* (`paper.tex`)

6 November 2026

Setup

Twelve runs: three models (Haiku 5.5, Sonnet 5.5, Opus 5.5) × two effort levels (low, high) × two repeats. Each run wrote a `report.tex` giving (1) the principal achievement, (2) the principal drawback and (3) the most worthwhile follow-up. I put each answer into a category based on what the report presents as its *headline* choice. Points raised only as supporting detail are counted separately in Table 3.

1. Principal achievement

All 12 runs chose the same principal achievement.

- **A1 – Unified BPS-limit framework (12/12).** The BPS decoupling limit that defines matrix theory (MpT), when applied at the asymptotic infinity of a Dp-brane geometry, *is* the near-horizon limit. This places matrix theory, AdS/CFT and IMSY, non-Lorentzian holography and the DLCQ^n/DLCQ^m classification in one scheme.

Every run also named the p-brane $T\bar{T}$ flow equations as the main concrete result. Most runs treated them as a by-product. Two Haiku runs (`high_1`, `low_2`) gave them equal billing as a second headline achievement. Two runs checked the flow equations numerically: Haiku `high_2` ($p = 1, 2$) and Opus `high_2` ($p = 0, 2, 3, 4, 5$, up to 6 scalars). Both found that the equations hold.

2. Principal drawback

- **D1 – Kinematic or conjectural only, with no dynamical test (8/12).** The evidence consists of matching limits and scalings. No observable (spectrum, correlator, thermodynamics, symmetry algebra) is compared across any new duality, so the DLCQ^n/DLCQ^m conjecture works as a classification rather than a falsifiable claim.
- **D2 – Unverified non-Lorentzian bulk backgrounds (3/12).** The headline weakness is that the new geometries are not shown to solve the MpT supergravity equations and torsion constraints, and that they rely on an ad hoc rescaling of ℓ (that is, of N).
- **D3 – The "intrinsic" $T\bar{T}$ generation of the bulk is not derived (1/12).** $\Omega(r)$ is inserted by hand from the known solution instead of following from the flow.

These categories overlap heavily. Every report mentions the unverified backgrounds somewhere (Table 3). The difference lies in which point the report puts first.

3. Recommended follow-up

- **F1 – Formulate or use MpT supergravity, verify the non-Lorentzian backgrounds and compute their isometry and super-algebras (7/12).** Variants differ in the starting case ($p = 0$, $p = 0, 1$, or M2) and in whether the supergravity equations must first be derived.
- **F2 – Non-abelian and/or supersymmetric extension of the p-brane $T\bar{T}$ flows (2/12).** Test against the Myers action, BFSS/BMN, or BLG/ABJM.
- **F3 – Dynamical holographic test of a specific new duality (2/12).** Opus `high_1`: DLCQ^1/DLCQ^2 via the M5 uplift ($AdS_7 \times S^4$ with a null circle), matching symmetries, sources and the protected spectrum. Opus `high_2`: large-N thermodynamics of the $p = 2$ (ABJM) flow compared with non-extremal M2-branes in asymptotically flat space.

- **F4 – Derive $\Omega(r)$ from the non-Lorentzian equations of motion (M2 case) (1/12).** This tests the intrinsic-bulk claim.

Breakdown by model and effort

Table 1: Headline choice per run. A1 = unified BPS/near-horizon framework. Drawbacks: D1 = no dynamical test, D2 = unverified backgrounds, D3 = $\Omega(r)$ put in by hand. Follow-ups: F1 = MpT supergravity check, F2 = non-abelian/susy flows, F3 = dynamical holographic test, F4 = derive $\Omega(r)$. The runner-up is the alternative that the report ranked second, if it named one.

Model	Effort/run	Ach.	Drawback	Follow-up	Follow-up detail	Runner-up
Haiku	low_1	A1	D2	F1	Start at $p = 0$, then $p \leq 3$	Susy abelian $T\bar{T}$ flow
Haiku	low_2	A1	D2	F2	Non-abelian flow vs. Myers/BFSS/BMN	Supergravity check (F1)
Haiku	high_1	A1	D2	F1	CAS check + isometry algebra	Derive $\Omega(r)$ from flow (F4)
Haiku	high_2	A1	D3	F4	$p = 2$, $\Omega(r)$ unknown in EOM	none ranked
Sonnet	low_1	A1	D1	F1	Formulate EOM, verify, symmetries	Non-abelian/susy flows (F2)
Sonnet	low_2	A1	D1	F2	M2/D2 flow vs. BLG/ABJM, Myers	Supergravity / observables
Sonnet	high_1	A1	D1	F1	Build sugra ($p = 0, 1$), then observables	Non-abelian flows (F2)
Sonnet	high_2	A1	D1	F1	Sugra + isometries + one observable	Non-abelian flows (F2)
Opus	low_1	A1	D1	F1	Expand sugra ($p = 0$ or 2), superalgebra	Non-abelian/susy 3d flow (F2)
Opus	low_2	A1	D1	F1	M2T sugra, superalgebra match	Non-abelian/susy flows (F2)
Opus	high_1	A1	D1	F3	M0T/D4 via M5 uplift: symmetries, spectrum	Quantise $p = 2$ flow
Opus	high_2	A1	D1	F3	ABJM $T\bar{T}$ thermodynamics vs. M2	none ranked

Table 2: Counts of headline choices by model and effort (two runs per cell).

	Haiku		Sonnet		Opus		
Category	low	high	low	high	low	high	Total
A1 Unified BPS/near-horizon framework	2	2	2	2	2	2	12
D1 No dynamical test / kinematic only	0	0	2	2	2	2	8
D2 Unverified non-Lorentzian backgrounds	2	1	0	0	0	0	3
D3 $\Omega(r)$ inserted by hand	0	1	0	0	0	0	1
F1 MpT supergravity check + symmetries	1	1	1	2	2	0	7
F2 Non-abelian / susy $T\bar{T}$ flows	1	0	1	0	0	0	2
F3 Dynamical holographic test	0	0	0	0	0	2	2
F4 Derive $\Omega(r)$ from EOM	0	1	0	0	0	0	1

Observations

- **Achievement:** all runs agree, regardless of model or effort.
- **Drawback by model:** every Sonnet and Opus run framed the main weakness as the lack of any *dynamical* test. Haiku framed it more narrowly, as specific unverified steps (the backgrounds, or the $\Omega(r)$ input).

Table 3: Specific weaknesses mentioned anywhere in the drawback section (not only as the headline).

	Haiku		Sonnet		Opus		
Weakness	low	high	low	high	low	high	Total
Backgrounds not shown to solve MpT sugra	2	2	2	2	2	2	12
Flows only classical/bosonic/abelian	2	2	2	2	2	2	12
Ad hoc rescaling of ℓ (i.e. of N)	2	0	2	2	2	2	10
No observable / quantitative test	1	1	2	2	2	2	10
Strong coupling / quantum effects at infinity, $p > 3$	2	1	2	2	2	0	9
Poincaré patch only	1	0	2	1	1	0	5
$\Omega(r)$ put in by hand (circularity)	0	2	0	0	0	2	4
Paper excessively long	0	0	0	1	0	1	2

- **Follow-up:** the modal choice, verifying the non-Lorentzian backgrounds in MpT supergravity (F1), is the "safe" option. It appears in every model. The non-abelian $T\bar{T}$ extension was the most common runner-up (6 runs) and was the main pick in 2 low-effort runs. Only Opus at high effort proposed a dynamical holographic test (F3). Both of those proposals were the most detailed and specific in the set (M5 uplift; ABJM thermodynamics with a $T_{\max}$ comparison).
- **Effort:** higher effort produced longer reports (Opus high: 130–166 lines versus 19–35 at low) and more independent checking, such as numerical verification of the flow equations. It changed the follow-up choice mainly for Opus.